

13. Analysis D: minimum detectable size at 50%, 10%, and 5%, with a predicted 0% point

D₅₀, D₁₀, and D₅ come from a psychometric curve fitted to the detection trials, after correcting for false alarms. The curve is fitted in $\ln D_{px}$, pooled over the (feature, station) pairs of the Z sweep. D₅ is the lowest measured point; the extended trials at the far stations are sized for it. The 0% point is not measured: a smooth curve never reaches zero, and no finite number of trials can prove a probability is zero. D₀ is therefore reported as an extrapolation of the fitted curves to the level DETECTION_ZERO_PREDICTION_LEVEL, flagged as a prediction, and cross-checked against a model with a hard floor and, for cutouts, against the geometric limit.

1. Independence check. For each feature, compute the lag-1 autocorrelation of its detected/missed sequence in acquisition order. It should lie within $\pm 2/\sqrt{n}$. Also check that outcomes at neighboring features in the same frame are uncorrelated. Correlation at either level means the offsets were too small or the features too close; widen PHASE_JITTER_SPAN_PX or FEATURE_ISOLATION_PX and repeat.
2. Detection rule. In the feature window (radius $D_{px}/2 + \text{DETECTION_WINDOW_MARGIN_PX}$), compute the deviation from the reference plane. For a disk, $\Delta = Z_{\text{back}} - Z$ (toward the sensor). For a cutout, $\Delta = Z - Z_{\text{front}}$ (away from the sensor). Candidates are valid reads with $\Delta > \tau$. The feature counts as detected if at least DETECTION_MIN_CONNECTED_PX connected candidates exist. No-reads do not count in the primary rule. A second, "no-read-inclusive" rule treats no-reads as candidates for cutouts, since a downstream reader might take a hole of no-reads to be a hole. Report both, because the gap between them is a direct measure of boundary bias (Section 14).
3. Threshold. For each Z, configuration, and window size, apply the rule to the blank sites. Set τ so that the false-alarm fraction equals DETECTION_FALSE_ALARM_TARGET. Then record the measured false-alarm rate γ with its Clopper–Pearson interval.
4. Psychometric fit. Model the raw detection probability as below, with $x = \ln D_{px}$, pooled over the features and stations. F is the logistic function, γ is fixed from the blank sites at each station, and the lapse rate λ is free but bounded to $[0, \text{DETECTION_LAPSE_RATE_MAX}]$. Fit α , β , λ by maximum binomial likelihood. The false-alarm-corrected probability is $P^*(D) = F$.

$$\psi(D) = \gamma + (1 - \gamma - \lambda) F\left(\frac{\ln D - \alpha}{\beta}\right), \quad D_{50} = e^{\alpha}, \quad D_{10} = e^{\alpha - \beta \ln 9}$$

5. Model dependence of D₁₀. D₁₀ sits in the tail, so it depends on the curve shape more than D₅₀ does. Fit three shapes for F: logistic, cumulative normal, and Weibull in $\ln D$. Report D₁₀

from the best fit by deviance, and the range across the three as model uncertainty. As a check that needs no model, fit a monotone curve to the corrected proportions with isotonic regression and interpolate the 0.1 and 0.5 crossings.

6. D_5, the lowest measured point. At each level, compute the two-sided Clopper–Pearson interval of the raw detection rate at CONFIDENCE_LEVEL and correct it for false alarms. Read D_5 from the fitted curve at the corrected probability of 5% (DETECTION_LOW_PROBABILITY), with its bootstrap interval, and report beside it the measured corrected rates and intervals of the levels nearest to it, so that the reader sees the point is measured, not extrapolated.
7. D_0, predicted (flagged). Extrapolate each fitted shape (logistic, cumulative normal, and Weibull in $\ln D_{px}$) below D_5 to the level DETECTION_ZERO_PREDICTION_LEVEL and report the size where it crosses; give the range across the three shapes as the model uncertainty, and a profile-likelihood interval from the threshold model, a Weibull with a location parameter ($P^* = 0$ for $D \leq D_0$). Label every reported D_0 as a prediction from the fitted curve, never as a measurement; the measured curve ends at D_5.
8. D_0, geometric limit (cutouts). Find the diameter at which A_geo from Section 12, Step 7 reaches zero, meaning no part of the back plate is seen by all the views a read needs. Below it, the primary rule cannot detect a cutout on any sensor of this baseline. A predicted D_0 well above the geometric limit means the sensor's processing, not geometry, sets the floor.
9. Confidence intervals. Bootstrap over poses (BOOTSTRAP_RESAMPLES), re-fitting γ , τ , and the curve each time. This gives intervals for D_50, D_10, D_5, and the predicted D_0.
10. Units, scaling, and Z dependence. Report each minimum in three forms: D_px, $\theta = D_{px} / f_x$ in mrad, and D in mm at each station. The Z sweep makes the noise covariate explicit: fit a logistic regression on $\ln D_{px}$ and $\ln \sigma_{tot}(Z)$ from Section 10 together. A negligible σ_{tot} term means the minimum is set by pixel or projector-dot sampling; a significant one means it is set by noise. Apply the scaling test of Section 12 here too: the detection curves of neighboring features must agree over their shared D_px range, and a disagreement is reported with its σ_{tot} attribution.
11. Outputs. D_detect_summary.csv with one row per configuration and station, plus one pooled row per configuration. Columns: γ , τ , D_50, D_10, D_5, the predicted D_0 with its model range, geometric D_0 (cutouts), and confidence intervals, each in px, mrad, and mm, for both detection rules. Figures: psychometric curves in D_px with binomial error bars, D_5, D_10, and D_50 marked and the predicted D_0 drawn as a dashed extrapolation, one trace per feature; θ vs Z for each minimum.